

Project

1 Ground Rules

This project gives you an opportunity to demonstrate your critical thinking, problem-solving, communication, technical skills and competencies. Either option will also result in an online document that you can link to in your resume or share as a writing sample with summer research programs, future employers, or graduate schools.

1.1 Groups

You are allowed to work in groups as large as two.

1.2 Submitting Work

You should publish your portfolio using GitHub Pages. There are three ways to do this (see the Quarto page [here](#)). If you want to use GitHub actions, you will find the video tutorial [here](#) helpful.

Remark: While not necessary, automating GitHub actions is a cool line in your resume and something cool to talk about in interviews.

You will submit a link as your final product. Everything should be viewable for grading, but I may reach out for code or GitHub repository access if I need more details.

1.3 Generative AI

Use your favorite flavor of generative AI as a peer, not as a ghostwriter. If you can outsource your learning, your potential employer can outsource your position. Use AI in a collaborative way to sharpen your thinking and communication, or to help you past a barrier, but the final product must be your own.

You may use generative AI for the following:

- Google Replacement: “Can you remind me of the difference between a fixed effect and a random effect?”
- Simplifying: “Can you help me understand [this topic] to me using the Feynman technique?”
- Completeness: “Here is a list of conditions for using Poisson regression. Are these correct? Am I missing any?”
- Unveiling R: “Can you produce code to fit a Poisson regression like the `glm()` function in R so that I can understand its inner workings better?”
- Code Troubleshooting: “I am not getting the same standard errors as R’s `glm()` function. Can you help me debug my function?”
- Clarity Check: “Here is my explanation of an offset. Is this clear for an introductory to intermediate statistics student?”
- Added Clarity: “What is the most common mistake students make when learning [this topic] or conducting [this analysis]?”
- Grammar: AI tools like Grammarly may be used to identify grammatical errors in your own writing, but it should not be used to rewrite your prose into a different voice.

AI can be very helpful as a critic. Try asking the following questions:

- What are potential counter-arguments to this excerpt? Are there alternative or complementary arguments that were not considered?
- Does this flow logically from the beginning to the end?
- Critically analyze this draft and identify areas where transitions between ideas, paragraphs, or sections could be strengthened.

- Where can the clarity, conciseness, or coherence in the writing be improved?
- Based on this excerpt, provide a summary of the main points. I'd like to see if they match my intended purpose.
- What is wrong with this analysis?
- Is my writing on this topic correct, complete, and free of errors?

Note: Carefully consider the back-and-forth around these questions. If you accept all feedback, you are outsourcing too much of your efforts. Aim to consider all feedback, but only implement those you independently agree with (roughly 30-60% of suggestions, depending on context).

You may not use generative AI for the following:

- Writing: Do not generate sentences or paragraphs of text for your portfolio.
- Analyses: You should not outsource any data analyses for your portfolio. Remember, your AI may not provide a thorough, complete, and accurate analysis of the data. It can also give you a correct answer that is not the correct answer in the context of our class.
- Reflection: Reflections on learning must be 100% your own. These are a record of your internal change, and if we outsource that, what are we even doing anymore?

Just as we did in the interviews, if I ask you to explain a specific derivation or a line of code in your work, you should be able to do so. If you cannot explain the “why” behind a piece of your portfolio, it can not count as “evidence of learning.”

AI Appendix

To maintain transparency, include an AI Appendix with subsections matching the sections of your portfolio. In each, include a bulleted list for each task you engaged with AI about. For example:

Prompt: I used Gemini to brainstorm what should be on the do/not do lists with AI.

Outcome: I added two additional points to the lists. The ideas are from Gemini, but I reworded them to be broader and in my own voice.

- What is the most common mistake students make when learning [this topic] or conducting [this analysis]?
- The part about not allowing AI to generate sentences or paragraphs of text.

2 Option 1: A Statistics Portfolio

2.1 What To Do

For each unit of the course, your portfolio should include important definitions, derivations, and modelling procedures that we will discuss this semester.

**** Unit 1:** Review of OLS Regression (HW1) **** Unit 2:** Mixed Effects Models (HW 2-3) **** Unit 3:** GLMs: Count Regression (HW 3-4) **** Unit 4:** GLMs: Logistic, Multinomial, and Ordinal Regression (HW 5-6) **** Unit 5:** Statistical Learning: Support Vector Machines, Random Forests, etc. (HW 6+)

You can view *Medium* (example [here](#)) or *Towards Data Science* (example [here](#)) articles for examples of this type of work, noting I expect yours to be at least as good as the best articles published in those venues. Each section should include simulated and real data analyses to demonstrate (1) a full understanding of the content *and* (2) an ability to synthesize and explain the information in an accessible way.

Remark: If I remove all “copied-and-pasted” or slightly rewritten content, there must be enough evidence of learning left over to illustrate the journey toward understanding of the course content.

For each unit, include revised solutions to the homework questions that correspond to that unit’s content. Again, this is to demonstrate your learning and re-learning of the material. You should resolve any comments I’ve made on your marked assignments *and* expand on your solutions as you see fit – noting that you have continued to learn after submitting those assignments. Please include reflections on this work, including a short description of changes since your initial submission.

Once your unit-level work is completed, there are two more tasks:

1. You should include one additional data analysis or simulation in which you either (1) reproduce the results in a publication or (2) perform a rigorous simulation study to answer a substantive question you have about the material. You can find articles with published data at [PLOS One](#), [Nature](#), and other journals.

2. You should include a reflection about the processes involved in creating this portfolio and what you learned as a result. If you're having trouble reflecting, consider Fink's Taxonomy of Significant Learning (Google Fink taxonomy aligned self-reflection questions).

2.2 Group Work Details

If you're working in a group, each of you is responsible for your own portfolio. Your group member and you should engage in frequent and substantive peer review. Ask questions, where do you get lost in the explanations or analysis? What polishing can be done to improve the portfolio?

3 Option 2: An Applied Analysis

3.1 What To Do

Vo (2018) conduct an analysis of 311 calls in New York City and population demographics. They used the Census API to determine whether certain census tracts are gentrified according to the methodology described [here](#), then they conducted a t -test to show that the average number of 311 calls was higher in gentrified neighborhoods than non-gentrified neighborhoods.

Hint: Vo (2018) have published all of their work (primarily `Python`) in a GitHub repository [here](#). What excellent data journalists!

In class, we discussed work by Dahir et al. (2025), who modeled the number of surveillance cameras based on population demographics across ten major U.S. cities. They used Poisson and zero-inflated Poisson regression to show that, while many expected cameras to be most common in predominantly Black areas, they are actually most frequent in neighborhoods with high racial diversity.

Hint: Dahir et al. (2025) have published all of their work (primarily `R`) in a GitHub repository [here](#). What excellent quantitative researchers!

Your first job is to create a Quarto website that reproduces the results in both projects. Note that you should conduct more rigorous checks of the conditions for use than either article does. This may or may not lead you to reconsider the analysis choices of either paper – e.g., should something other than a t -test be used in Vo (2018)? Should negative binomial regression be used in Dahir et al. (2025)?

Your second job is to assess the findings across papers. That is, does the gentrified result hold in the outcome explored in Dahir et al. (2025)? Does the unexpected population demographic relationship hold for the outcome explored in Vo (2018)? You might also consider other confounders available from the United States Census.

Remember, you can collect data from the United States Census using the `tidycensus` package for `R` (Walker and Herman 2026). Further, the 311 call database is available for download [here](#). Note that the data are split into two files from 2010-2019 and 2020 to the present day.

Note: The NYC 311 dataset is extremely large. You will need to filter by specific complaint types as noted in Vo (2018). Save the resulting dataset and work from there instead of reloading that massive file repeatedly.

3.2 Generative AI details

Generative AI can be particularly helpful here. If you are having trouble understanding the `Python` code, ask your favorite Generative AI to translate a code block from `Python` to `R`. You might also find it helpful to ask it to add comments to help you understand the code better.

3.3 Group Work Details

If you're working in a group, you will submit one project write-up. Each group member should create a substantial portion of the writing *and* the code. Group members should engage in frequent and substantive peer review of each other's work.

You should invite me to your GitHub and/or Overleaf repository so that I can track the progress and contributions of both group members. If y'all work on one computer in a joint-work session, the contributions should be clearly denoted in the commit details.

Note: I would recommend creating two GitHub pages in the end so that you each have a unique link for sharing. You can add a tagline saying something to the effect of “By [first1] [last1] (created in collaboration with [first2] [last2]).”

4 Option 3: Software Development for Statistics

4.1 What To Do

Create a statistical software toolkit that implements the methods we’ve discussed in class. The toolkit should be available as either (1) a functional R Package or (2) an Interactive Shiny Dashboard that automates count regression.

Note that our count regression unit was expansive including Poisson Regression, Quasi-Poisson, Negative Binomial Regression, and zero-inflated models. Your software should help users complete the following tasks in a manner that a statistician would:

1. Summarize data for a count regression model.
2. Fit the types of count regression models we discussed in class, starting with Poisson regression.
3. Assess the conditions for using Poisson regression and suggest appropriate alternatives when necessary.
4. Provide final model validation, including all necessary diagnostic plots, statistics, and inferential tests.
5. Interprets all model output in clear terms (e.g., not on link scales, etc.)

Your software toolkit should include the following deliverables:

- Documentation: All tasks and output should be well document. You can do this in the user interface if you are building an Shiny app, and you might find the `roxygen2` package (Wickham et al. 2025) helpful for documenting package functions if writing an R package.
- Vignette: A Quarto page where you use your software to analyze the datasets from class.
- Unit Testing: Write scripts – either R scripts or click-by-click scripts – for testing the software.

When writing an R package, you might consider using the `testthat` package (Wickham 2011) to define quality analysis goals that can be automatically checked using the `test()` function from the `devtools` package (Wickham et al. 2026). Alternatively, for a Shiny app, you can use the `shinytest2` package (Schloerke 2026) to automate the click-by-click instructions.

The datasets from class should can make up a portion of unit-tests, but also include “bad” data like NAs or NaN inputs. Your software should handle these cases and clearly inform the user of what is done with those observations.

Once your software toolkit and documentation is done, choose one task below:

- Expand the functionality of your toolkit to support mixed effects versions of the count regression models.
- Add the capability to fit at least one of (1) generalized Poisson, (2) compois, or (3) Tweedie regression models.

Make any necessary additions to your documentation, and vignette. If you added a new model, find a paper with published data at [PLOS One](#), [Nature](#), or another journal that requires publicly posting data.

4.2 Generative AI details

For this option, generative AI is a powerful tool. You can use it to help you remember certain functions and their syntax or to debug specific error messages you come across. However, as noted in the Ground Rules, the logic of the statistical tests and the interpretation of the outputs must be yours.

4.3 Group Work Details

If you’re working in a group, you will submit one piece of software and accompanying documentation. While it may be natural to split on backend/front-end tasks or mathematical/programming tasks, each group member should create a substantial portion of both the writing *and* the code. Group members should engage in frequent and substantive peer review of each other’s work.

You should invite me to your GitHub repository so that I can track the progress and contributions of both group members. If y’all work on one computer in a joint-work session, the contributions should be clearly denoted in the commit details.

Share your vignette using the GitHub pages method described in the Ground Rules. Even though I will have access through your shared GitHub repository, make sure your R package or app are available to your readers.

Note: I would recommend creating two GitHub pages in the end so that you each have a unique link for sharing. You can add a tagline saying something to the effect of “By [first1] [last1] (created in collaboration with [first2] [last2]).”

References

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